



Experimental Investigations of a Dual-Mode Skin-Actuated-Camber with Embedded Twist (SACET) Morphing Wing

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A dual mode Skin Actuated Camber with Embedded Twist morphing wing is developed to approximate a bell-shaped lift distribution and the associated tip vortex free wake. Three morphing concepts are conceived for a rectangular wing using simple geometric models that link actuator motion to local rib twist. The concepts use independent servo driven ribs, a gear train that drives multiple ribs with a single shaft, and a linkage based single rod layout under a flexible skin. One configuration with embedded rib servos and TPU skin is fabricated and tested in the University of Dayton Low Speed Wind Tunnel and is compared with rigid wings that has the target twist and camber distributions. Measured lift, drag, and quarter chord moment coefficients show that the current prototype does not achieve the commanded twist, because the twist only morphing configuration behaves like the untwisted wing, while the rigid twisted wing exhibits the expected changes in lift, drag, and moment. The results indicate that higher twist authority and reduced skin resistance are required before the full aerodynamic benefits of dual mode morphing for bell shaped loading can be realized.

Nomenclature

a	= Lift Curve Slope.	q	= Dynamic Pressure
AR	= Aspect Ratio	Re	= Reynolds Number
b	= Wingspan	S_{ref}	= Reference area
$\bar{c}$	= Mean Aerodynamic Chord	U_{∞}	= Freestream Velocity
C_D	= Drag Coefficient, $C_D = \frac{D}{qS_{Ref}}$	y_b	= Normalized Spanwise Location
C_L	= Lift Coefficient, $C_L = \frac{L}{qS_{Ref}}$	α_{Root}	= Angle of Attack at Root Chord
D	= Drag force	$\theta(y_b)$	= Wing Twist Distribution
L	= Lift force	ρ	= Air density
d_p	= Pitch Diameter	r_p	= Pitch Radius
z	= Number of Teeth	m	= Module
ϕ_p	= Arc Length	x_{rack}	= Displacement of Rack
L	= Distance	s_{rib}	= Attachment Points
c_0	= Geometric Offset	θ	= Rib Twis

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I. Introduction

The concept of morphing aircraft has attracted sustained interest in recent decades, offering a means to dynamically tailor an aircraft's aerodynamic shape to improve efficiency, controllability, and mission adaptability [1]. Morphing techniques encompass a range of geometrical changes, including span extension, wing sweep, chordwise camber control, and spanwise twist morphing. While many UAV morphing designs have concentrated on camber variation via trailing edge deflection or compliant skins [2], spanwise twist morphing remains comparatively underexplored despite its potential to provide continuous roll control, gust load alleviation, and trim adjustment without reliance on discrete control surfaces. The concept of spanwise twist itself is not new. The Wright Flyer relied on mechanical warping of its wings to initiate banked turns, well before ailerons became standard [3]. However, the reintroduction of twist morphing into modern unmanned aerial vehicles (UAVs) presents new engineering challenges and opportunities, particularly in terms of actuation methods, structural compliance, and wing surface continuity. Unlike discrete control surfaces, which introduce sharp geometric discontinuities and hinge-line flow separation, twist morphing can modulate the spanwise lift distribution more gradually, resulting in smoother aerodynamic response and lower induced drag penalties [4].

Recent UAV-scale demonstrations have shown that distributed twist actuation is feasible with lightweight composite structures and servo-driven mechanisms. Schlup et al. [5] presented the XM-2 UAV, which integrates a central wing segment capable of $\pm 15^\circ$ of twist via a foam-core structure and elastomeric skin. The twist, actuated by a combination of root-mounted servos and flexible ribs, enabled the vehicle to maneuver in roll without the need for ailerons. The design maintained a smooth aerodynamic contour and demonstrated strong correlation between commanded twist and aerodynamic moment generation. Building on these foundational efforts, Gatto [6] implemented an active twist control (ATC) mechanism using a rotating torque tube embedded within a modular UAV wing. Spanwise rib rotation was driven by servos housed at the root and mid-span. The design featured closed-loop control using onboard sensors, and wind tunnel results indicated a substantial lift-to-drag improvement up to 72% in certain flight regimes relative to fixed-wing baselines. These results highlighted the aerodynamic effectiveness of real-time morphing and the ability to dynamically optimize wing configuration under varying conditions.

More compact and scalable approaches have emerged to address the limitations of internal volume and mechanical complexity. Luo et al. [7] introduced a shaft-based twisting architecture employing segment gears installed at each rib, allowing for distributed twist deformation while maintaining geometric simplicity. The segment gear system facilitated spatial decoupling between adjacent ribs and enabled large twist angles to be achieved in a constrained internal envelope. This approach, although initially designed for biomimetic robots, is readily adaptable to UAV platforms with spanwise morphing capabilities.

Maintaining surface smoothness during morphing remains a central design challenge. Many prior twist morphing designs suffered from surface discontinuities or buckling, particularly when deforming thin airfoils under load. Zhu et al. [3] surveyed recent advancements in morphing skins and highlighted the role of compliant elastomeric materials and additively manufactured lattice structures in resolving this issue. These flexible skins can undergo significant spanwise strain while preserving chordwise stiffness, enabling effective transmission of actuation torque to aerodynamic surfaces without localized separation. Smart materials have also been proposed for localized twist actuation. Piezoelectric macro-fiber composites (MFCs), shape memory alloys (SMAs), and dielectric elastomers have been explored as distributed actuation systems for both camber and twist morphing applications [8]. However, most implementations remain limited by actuation force, bandwidth, or complexity in feedback control. While promising for fine-scale deformations, such materials often require supplementary mechanisms to produce UAV-scale structural response.

Recent UAV prototypes have demonstrated the aerodynamic value of twist morphing. Negahban et al. [9] implemented twist morphing ailerons and winglets in the UAS-S45 platform using shape optimization to enhance roll control and reduce induced drag. Compared to conventional ailerons, the twist-enabled control surfaces increased rolling efficiency by 34% and reduced drag during cruise by up to 25%. Such performance gains highlight the viability of distributed twist morphing for both control and efficiency improvements.

A. Achieving Bell-Shaped Lift Distribution Through Morphing

Bell-shaped lift distributions have gained renewed interest as an alternative to the classical elliptical loading for high-aspect-ratio and flying-wing configurations. Compared to an elliptical distribution, a bell-shaped profile shifts lift inboard and allows small regions of reduced or even negative lift near the tips, which can simultaneously reduce induced drag, alleviate root bending loads, and mitigate adverse yaw tendencies. The impact of this non-elliptical loading on the wake topology is illustrated in Figure 1. The top row shows numerically simulated wakes at $x/c = 4$ for wings with uniform, elliptical, and bell-shaped lift distributions. Both the uniform and elliptical cases exhibit a strong rolled-up wingtip vortex, whereas the bell-shaped case produces no discernible tip vortex in the near wake. Instead, the vorticity is redistributed inboard into a weaker, flattened “pancake-like” structure. The bottom row of Figure 1 further validates this behavior using panel-method predictions, RANS simulations, and PIV measurements at $x/c = 4$, all of which confirm the absence of a concentrated wingtip vortex for the wing twisted to achieve a bell-shaped lift distribution [15]. Our prior experimental investigations on a rectangular wing with bell-shaped lift distribution [13] and on a scaled Prandtl-D wing [14][16] revealed similar results.

These results highlight that tailoring the spanwise loading toward a bell-shaped profile provides a means to control not only induced drag and bending moment, but also the topology and spatial footprint of trailing vorticity. This motivates morphing strategies that can reproduce this distribution in flight, especially in off-design conditions. A morphing wing capable of independently commanding spanwise twist and camber offers a flexible means to approximate such bell-shaped loading in flight and, at its design condition, achieve a wake that is effectively free of a concentrated tip vortex.

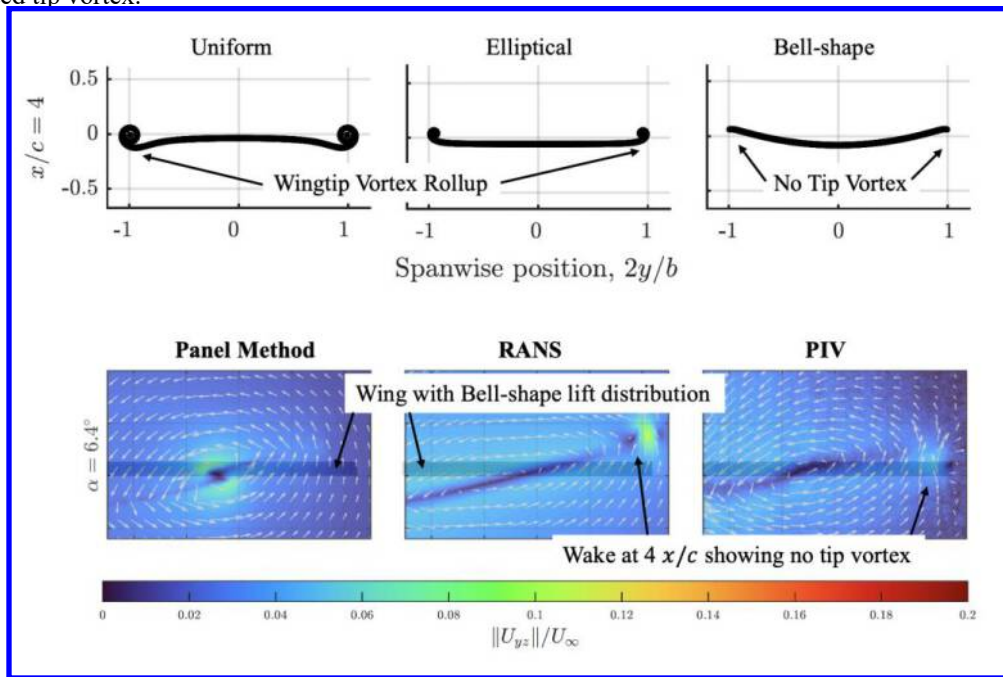


Figure 1 Wake topology at $x/c = 4$ for wings with uniform, elliptical, and bell-shaped lift distributions, showing elimination of the rolled-up wingtip vortex for the bell-shaped case. [15]

The coupling of spanwise twist with camber morphing therefore represents a new frontier in multi-functional wing design, particularly for tailoring spanwise lift distributions toward bell-shaped profiles and reproducing the associated tip-vortex-free wake at the design condition. The present work builds on prior work by Pabon et al. [10], who introduced a Fishbone Skin-Actuated-Camber (SAC) morphing wing that used spanwise-aligned skin tendons to modulate trailing-edge camber without hinge gaps or buckling, achieving flap-like performance with smoother surface pressure distributions and reduced drag at high deflection angles. Along with the same camber actuation method, the present work embeds a segment-linkage-based spanwise twist mechanism within the same framework to create a Skin-Actuated-Camber with Embedded Twist (SACET) morphing wing. The system uses a root-mounted shaft with a

connected linkage driven by a servo embedded within spanwise-distributed ribs, providing independent twist along the wing while preserving the original camber-morphing capability. Actuation is achieved with AGFRC B13DLM V2 servos and an Arduino board control architecture.

Wind tunnel experiments in the University of Dayton Low-Speed Wind Tunnel (UD-LSWT) are used to characterize and diagnose the aerodynamic response of the integrated camber twist system under varying angle of attack and twist inputs. Static force measurements are compared against baseline models with the fixed targeted lift profiles. The overarching motivation is to demonstrate a scalable, lightweight morphing architecture that can reshape the spanwise lift distribution toward bell-shaped loading and its favorable wake characteristics while maintaining low mechanical complexity suitable for experimental validation and eventual flight testing.

II. Morphing Wing Configurations

Three morphing-wing configurations were developed to approximate a bell-shaped lift distribution. For a 16 in. span, the required spanwise twist distribution at a design $C_L = 0.3$ is shown in Figure 2, with the local twist decreasing smoothly from approximately $+6^\circ$ near the root to about -2° at the tip.

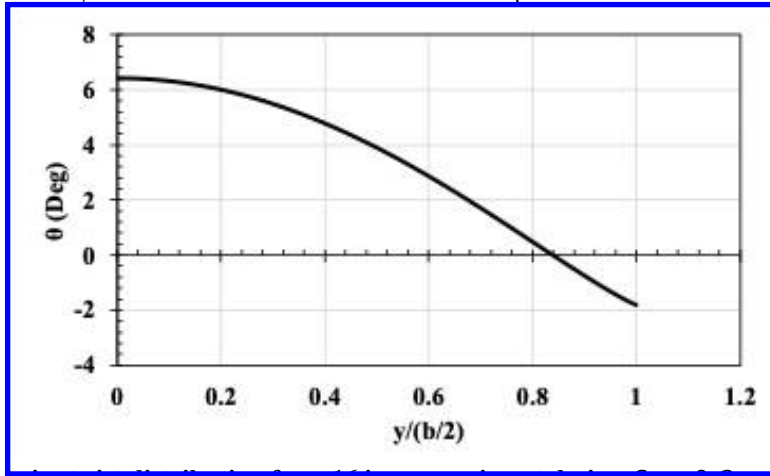


Figure 2 Target spanwise twist distribution for a 16 in. span wing at design $C_L = 0.3$, approximating the twist required for bell-shaped lift distribution.

To realize this distribution, three SACET morphing-wing designs were conceived, and each configuration is described in the following subsections.

A. SACET with Rib-Embedded Servos

This design consists of five main components. PLA ribs, TPU skin patches inserted between each rib, a central rod that spans the entire wing, an internal twisting mechanism that is inspired by Luo et al. [7], and an internal cambering mechanism that is based on Pabon et al. [10]. These elements combine to create the SACET morphing wing featuring ribs that can change both camber and twist. The wing has an 8 in. chord and a 16 in. span. A schematic of this configuration is shown in Figure 3. Thin strips of tape are applied along the trailing edge to maintain a smooth contour across the span. However, it is worth noting that the goal is to improve the design to negate the need for tape. As shown in Figure 3, the SACET wing is assembled from five PLA rib sections and four TPU skin patches, starting with a rib section at the root and alternating down to the sensor interface plate. All rib sections are rigidly attached to the central rod in the vertical direction but are allowed to slide laterally, which enables twist. Each TPU skin patch attaches to the neighboring ribs through individual spokes that latch into holes in the PLA ribs. The central rod is printed in two pieces and joined at midspan. The entire wing mounts to a sensor interface plate that is printed from PC.

Referring to Figure 4, the SACET rib and skin skeleton is adapted from the patented design in Pabon et al. [10]. The leading edge and spar consist of a rigid PLA structure, while the trailing edge remains flexible through the thin PLA skin geometry. A servo cavity is located at the outboard end of each rib. Each servo carries a custom PLA servo

horn that attaches to the skin and actuates camber morphing. A T-shape latch interface between the internal geometry and the skin helps maintain the airfoil shape while still allowing for deflection. Two steel rods provide a rigid connection between the servo and skin to transmit the motion required by the twisting gear mechanism. Multiple ribs are distributed within the skin, and each rib can be actuated independently to generate spanwise-varying camber.

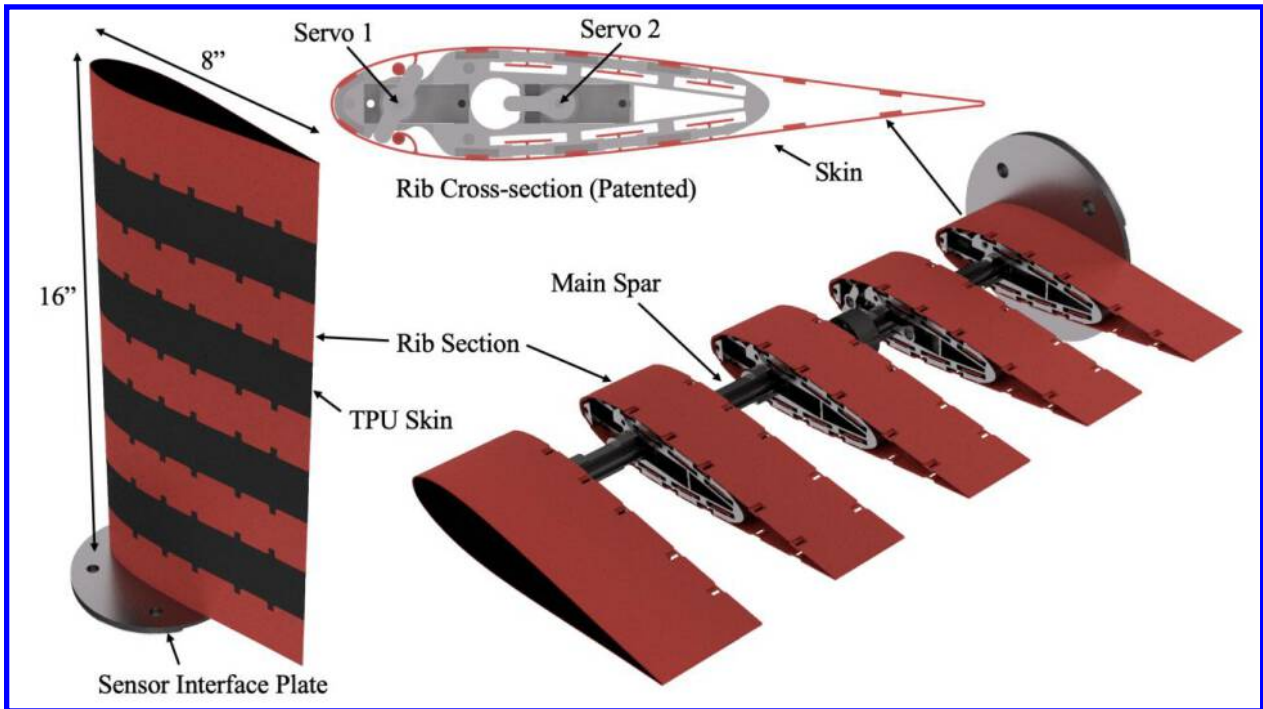


Figure 3 SACET morphing wing with independent servos, showing rib sections, TPU skin patches, main spar, and sensor interface plate for combined twist and camber morphing.

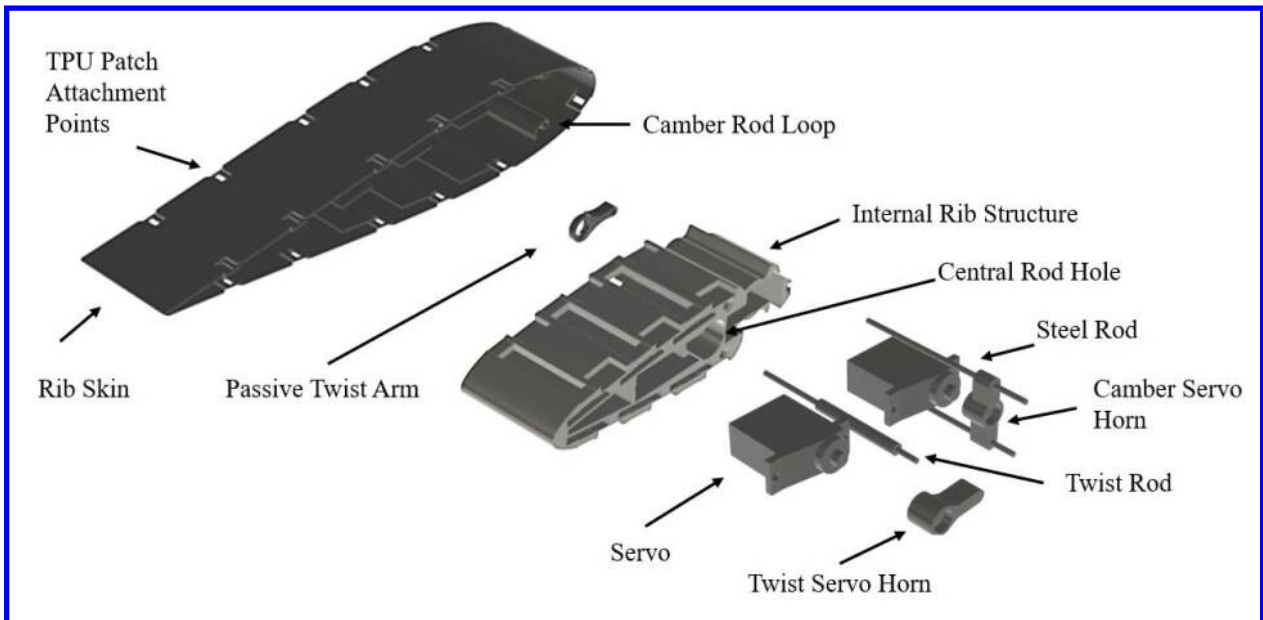


Figure 4 Exploded view of SACET rib and skin assembly, highlighting internal rib structure, camber and twist servos, steel alignment rods, and TPU patch attachment points.

The gear mechanism, inspired by Luo et al. [7], is shown in Figure 4 and consists of several interacting components. Each rib hosts its two servos with a unique set of 3D-printed servo horns. The twist horns engage with

the central rod such that the entire rib rotates about the main spar and produces local twist. To promote rigid contact with the central rod, twist along each rib, a small steel rod with a PLA sleeve is inserted just below the twist servo horn and also fits into the central rod. The magnitudes of twist and camber are commanded through an Arduino controller that runs custom code adapted from Pabon et al. [10]. This gear mechanism is designed to produce two twist cases in combination with two camber cases. The twist cases consist of a baseline configuration with no twist and a configuration that approximates the bell-shaped lift distribution. The camber cases consist of a baseline with no camber and a configuration with a prescribed positive camber deflection along the trailing edge. These combinations of twist and camber are implemented on the SACET morphing wing and compared with corresponding configurations on a rigid reference wing.

B. SACET with Gear Shaft

To reduce actuation complexity while still enabling spanwise twist, a second morphing concept was developed in which all ribs are driven by a single rotating gear shaft. Each rib contains an internal rack that is constrained to translate in the chordwise direction, and this rack is engaged by a sector gear mounted on a common geared shaft aligned with the main spar, as shown in Figure 5. The geared shaft is rotated by a single actuator at the wing root. As the shaft turns, each sector gear converts the shaft rotation into a linear translation of the corresponding rack. This translation, applied at an offset from the rib's torsional axis, produces a twist of the rib and delivers a smooth change in the wing twist. A flexible skin covers the mechanism and maintains an aerodynamically smooth outer surface.

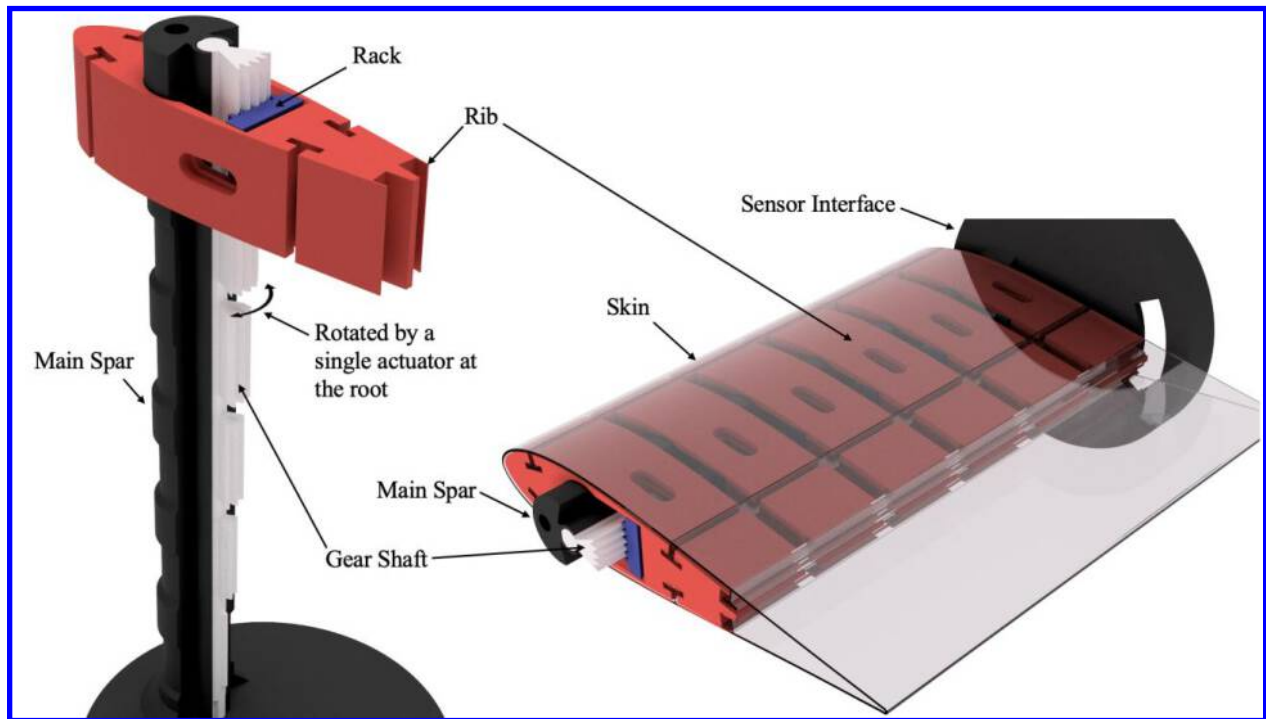


Figure 5 SACET gear train concept where one root actuator drives a geared shaft and racks to twist all ribs beneath a flexible skin.

The relationship between shaft rotation and rib twist can be described by a simple kinematic model. For a spur or sector gear with module m (mm) and number of teeth z , the pitch diameter and pitch radius are

$$r_p = \frac{d_p}{2} = \frac{mz}{2} \quad (1)$$

A small rotation of the geared shaft, and hence the pinion, by an angle $\Delta\phi_p$ produces an arc length at the pitch circle of

$$s = r_p \Delta\phi_p \quad (2)$$

Assuming pure rolling contact, this arc length is equal to the linear displacement of the rack,

$$\Delta x_{rack} = r_p \Delta\phi_p \quad (3)$$

The rack is attached to the rib at a distance L_c from the rib's twist axis (approximately coincident with the spar). For a small rib twist $\Delta\theta$, the attachment point moves along an arc

$$\Delta s_{rib} \approx L_c \Delta\theta \quad (4)$$

Neglecting compliance, the rack displacement and the rib-attachment arc length are equal, $\Delta x_{rack} = \Delta s_{rib}$, leading to

$$r_p \Delta\phi_p \approx L_c \Delta\theta \quad (5)$$

Solving for $\Delta\theta$,

$$\Delta\theta = \frac{r_p}{L_c} \Delta\phi_p \quad (6)$$

In the present design, the effective control-arm length L_c is modeled as the sum of a constant geometric offset c_0 and the pitch radius,

$$L_c = c_0 + r_p \quad (7)$$

where c_0 is the distance between the center of the main spar and the center of the gear shaft. Substituting this into the kinematic relation gives,

$$\Delta\theta = \frac{r_p}{c_0 + r_p} \Delta\phi_p \quad (8)$$

Expressed in degrees, the twist per degree of shaft rotation becomes

$$\frac{\Delta\theta_{deg}}{\Delta\phi_{deg}} = \frac{r_p}{c_0 + r_p} = \frac{mz/2}{c_0 + mz/2} \quad (9)$$

Thus, for a given actuator command Φ_{deg} , the resulting rib twist is approximately

$$\theta_{Deg} \approx \left(\frac{r_p}{c_0 + r_p} \right) \Phi_{deg} \quad (10)$$

under the small-angle assumption. This expression shows that the twist sensitivity depends only on the pitch radius (or equivalently, on the combination of module and tooth count) and on the fixed geometric offset c_0 .

Parametric Gear Design

To design a shaft that generates the desired spanwise twist distribution using a single actuator, the influence of gear geometry on twist sensitivity was examined through a parametric study. The gear module m was varied over a

range of standard values, and the number of teeth z was varied over a practical range compatible with the rib chord and thickness. For each pair (m, z) , the pitch radius $r_p(m, z) = mz/2$ and the corresponding twist per degree of shaft rotation were evaluated.

$$\theta_{deg}^i(m, z) \approx \left(\frac{r_p(m, z)}{c_0 + r_p(m, z)} \right) \quad (11)$$

Manufacturing and installation constraints impose upper and lower bounds on the gear size, which were modeled as allowable limits on the pitch radius. The gears must be large enough to be manufactured but also small enough to fit inside the wing.

$$r_{min} < r_p(m, z) < r_{max} \quad (12)$$

For each pair (m, z) , the corresponding pitch radius r_p and the twist angle θ generated by a 20° rotation of the geared shaft were computed and is shown in Figure 6. The contours represent the resulting rib twist θ (in degrees) for a 20° shaft rotation. The black curves indicate loci of constant pitch radius, with labels $r_{min} = 6.8$ mm and $r_{max} = 10$ mm. These two radii define the lower and upper bounds on the admissible gear size, imposed by manufacturing and packaging constraints. Thus, the feasible design region for the pinion geometry lies between the two contour lines. Within this band, a discrete combination of gear module and tooth count was selected to achieve the required twist level at each spanwise station while satisfying the geometric constraints of the mechanism.

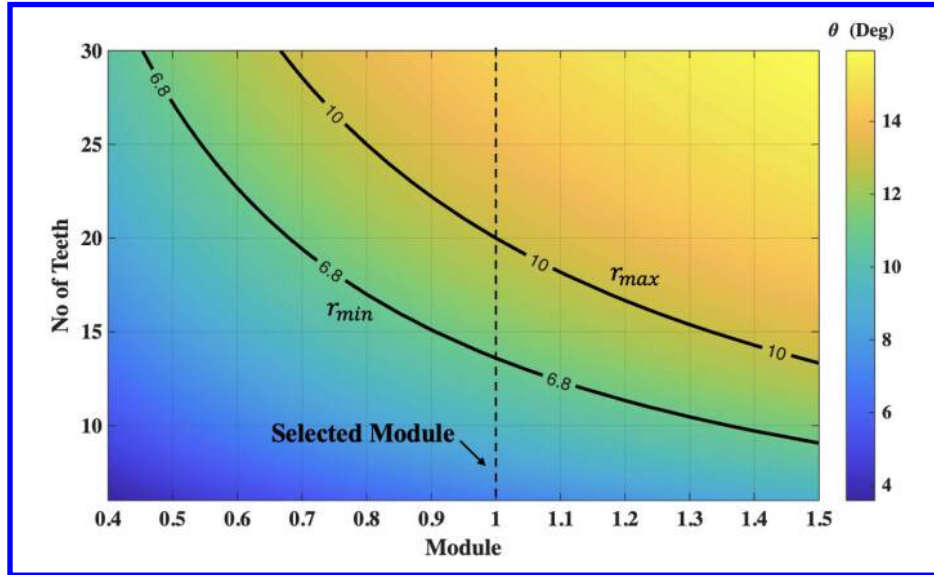


Figure 6 Design map for gear-train twist mechanism showing achievable rib twist angle versus gear module and tooth count, with minimum and maximum radius constraints.

C. SACET with Linkages

A third morphing concept was also developed to replace the rack-and-pinion shaft with a single actuation rod that runs spanwise parallel to the main spar as shown in Figure 7. Each rib is connected to this rod by a short in-plane linkage pinned at both ends. One pin is attached to a crank on the actuation rod, and the other is attached to the rib at a fixed chordwise offset from the main spar. As the actuation rod is rotated by a single servo at the root, all linkages are driven simultaneously. Because the linkage lengths vary from rib to rib, the resulting twist of each rib is different, which allows a prescribed spanwise twist (and therefore lift) distribution, in this case, the bell-shaped lift distribution to be generated with only one actuator. A continuous flexible skin is bonded to the rib tips, producing a smooth aerodynamic surface as the ribs rotate.

In a cross-sectional view, the mechanism is planar. The main spar defines the twist axis of each rib, and the actuation rod is located at a fixed center distance d from this axis. At rib station i , the linkage connects a crank of length a_i on the actuation rod to a pin on the rib. When the actuation rod is rotated by an angle ϕ (actuator command), this crank motion drives the rib pin, which in turn causes the rib to rotate by a twist angle θ_i about the spar.

The geometry can be idealized as a planar triangle formed by the spar axis, the actuation-rod axis, and the linkage joint. For a given rib station, the center distance d between the spar and the actuation rod (identical for all ribs), the commanded rotation of the actuation rod ϕ , and the desired rib twist θ_i at that station are known. The unknown is the crank (lever-arm) length a_i on the actuation rod that yields the required twist when the rod rotates by ϕ .

Let the rib twist θ_i be defined as the rotation of the rib about the spar such that the rib-side attachment point is displaced transversely by an amount

$$\Delta y_i = d \tan \theta_i \quad (13)$$

relative to the untwisted configuration (small-angle approximation in the spanwise direction). The motion of the actuation-rod crank of length a_i , rotated by ϕ , produces a transverse displacement of its tip that can be decomposed into components along and perpendicular to the line of centers. By enforcing compatibility between the displacement of the crank tip and the displacement of the corresponding pin on the rib (the two ends of the linkage must coincide), a simple trigonometric manipulation leads to the inverse kinematic relation for each rib station i .

$$a_i = \frac{d \tan \theta_i}{\sin \phi - \tan \phi \cos \phi} \quad (14)$$

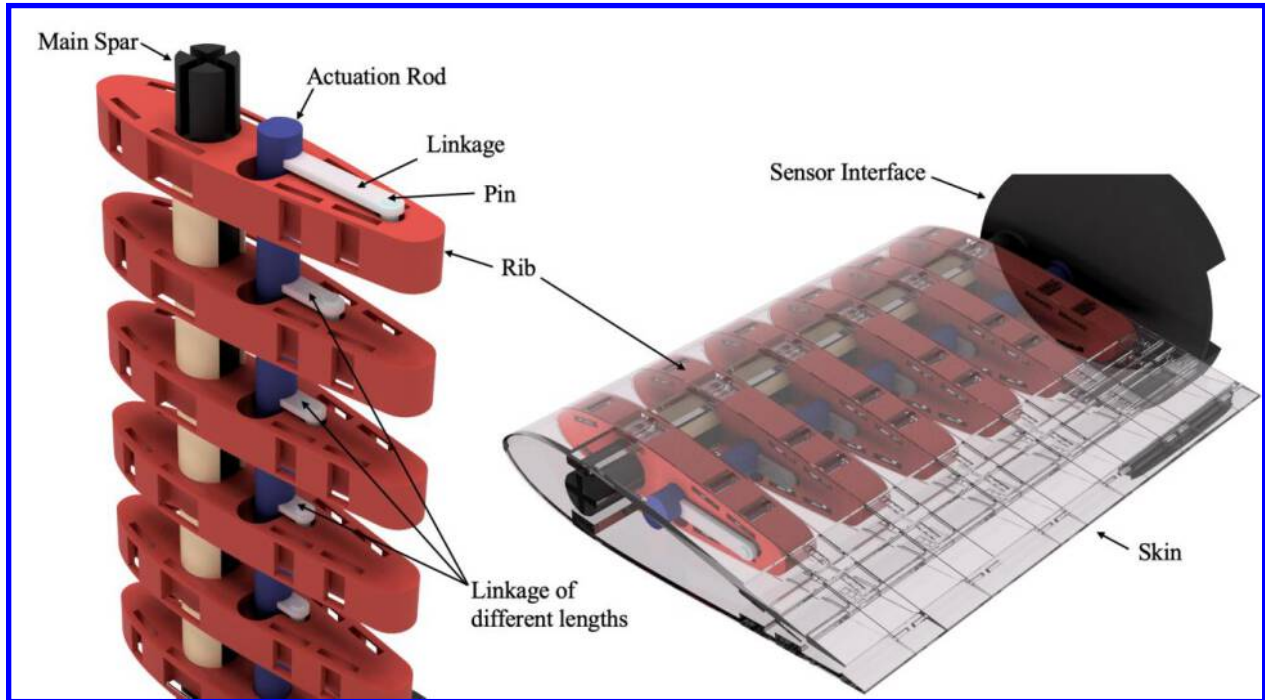


Figure 7 SACET linkage-based twist concept where a single actuation rod moves pins and linkages of varying length to generate prescribed spanwise twist beneath a flexible skin.

Equation (14) explicitly shows how the required lever-arm length depends on the chosen center distance d , the global actuation ϕ , and the desired local twist θ_i . For a given mechanism layout (d fixed) and a chosen actuator angle ϕ , the linkage lengths along the span are obtained simply by evaluating Equation 14 at each station. For the model shown in Figure 7, the center distance between the main spar and the actuation rod was fixed at

$d = 22.4$ mm by volume constraints inside the rib, and the maximum actuator rotation was chosen as $\phi = 14^\circ$. The target twist distribution θ_i along the span was selected to approximate a bell-shaped lift distribution, and discretized at six rib stations. The corresponding crank lengths are shown in Figure 8. As the magnitude of the required twist increases toward the tip, the lever-arm length grows rapidly. Equation (14) also reveals an important design trade-off. For fixed twist θ_i , decreasing the center distance d or the actuator angle ϕ increases the required arm length a_i . If the denominator in Equation 14 becomes small, the lever arms grow rapidly and may exceed the chord length of the wing. Conversely, increasing ϕ or d reduces the required linkage lengths but may conflict with other geometric or actuator-torque constraints. In practice, ϕ and d were selected to keep all a_i within manufacturable limits while still achieving the desired twist distribution.

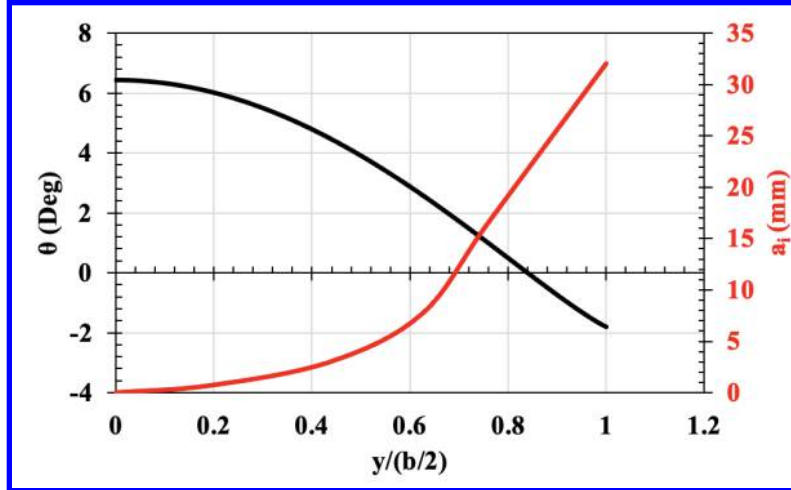


Figure 8 Target spanwise twist distribution and corresponding required linkage offset a_1 for the linkage-based SACET concept, plotted versus normalized span $y/(b/2)$.

All three designs were prototyped, but only the SACET wing with rib-embedded servos was tested in the wind tunnel. The experimental setup and the results will be discussed in the following sections.

III. Experimental setup

All experiments were performed in the University of Dayton Low-Speed Wind Tunnel (UD-LSWT) operating in its open-jet configuration. The facility features a 16:1 contraction ratio, six anti-turbulence screens, and a square test section measuring $0.762 \text{ m} \times 0.762 \text{ m}$. The effective test section length is 0.183 m. Downstream, a $1.128 \text{ m} \times 1.128 \text{ m}$ collector guides the expanded flow into the diffuser section. Freestream velocity was monitored using a pitot-static probe connected to a Siemens QBM-3230-U5D analog pressure transducer. The pressure transducer was calibrated using a TSI T600 micromanometer. All pressure data were acquired at the same sampling rate as the force/torque (F/T) sensor using a National Instruments NI-USB-6259 data acquisition unit.

A. Force Based Experimental Setup

A schematic of the experimental setup is shown in Figure 9. The morphing wing model was mounted vertically to a rotary stage and secured to an ATI Gamma F/T balance assembly. The F/T sensor provided six-axis force and moment data, with a calibrated uncertainty of 1.25% in the Y-axis (normal force) and 0.75% in the X-axis (axial force). A $0.914 \text{ m} \times 1.829 \text{ m}$ splitter plate was installed beneath the model to support symmetric boundary conditions, enabling half-span testing. A picture of the SACET wing with rib embedded servos installed in the wind tunnel is shown in Figure 10.



Figure 9. (a) Schematic of the experimental setup (b) Photograph of the experimental setup showing the morphing wing model mounted in the test section of the wind tunnel.

The test matrix, summarized in Table 1, covered two Reynolds number conditions to evaluate the aerodynamic sensitivity of the morphing wing across operational regimes. A freestream velocity of 20 m/s and 30 m/s corresponded to chord-based Reynolds numbers of approximately 270,000. The angle of attack (α) was varied from -5° to $+25^\circ$ in 1° increments for the rigid wings, and -5° to $+20^\circ$ for the morphing wing. This variation is due to caution being exhibited for the safety of the SACET wing and the wind tunnel at high angles of attack. Each test configuration was repeated twice to assess data repeatability. To minimize drift in the F/T sensor output, tare readings were recorded with the tunnel powered off immediately before and after each run. The average of the two tare values was subtracted from the respective load measurements. If the difference between the pre- and post-test tare values exceeded 0.1 N on any axis, the data for that run were discarded, and the test was repeated.

Five total wing configurations were tested; all wing configurations had an 8-inch chord length and a 16-inch span. The first two wings tested were rigid bodies, meaning they were not capable of actuation. These wings were used as a baseline to compare the performance of the SACET wing when actualized under similar conditions. Three conditions of the SACET wing were tested. One, with no twist and no camber actualized, one with just twist actualized, and one with both twist and camber actualized.

Table 1 Test Matrix for SACET with Rib Embedded Servo Morphing Wing Testing

Wing Type	Reference Area (m²)	Reynolds Number	Angle of attack (deg)	Freestream Velocity (m/s)
Rigid – Twist – No Camber	0.0826	270,000	-5 to 25	20
Rigid – Twist – Cambered				
SACET – No Twist – No Camber				
SACET – Twist – No Camber				
SACET – Twist – Camber				

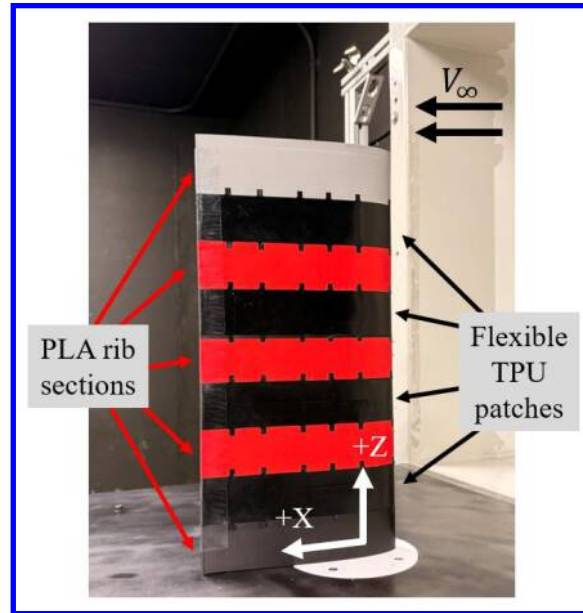


Figure 10. Fully assembled SACET Wing Mounted on Balance Assembly.

B. Baseline fixed model

To compare with the SACET wing, two “baseline” wings were created with the same geometry of an 8 in. chord, 16 in. span, and an Eppler 479 airfoil, shown in Figure 11. Both were printed out of PLA material in two pieces and joined together with screws and epoxy. Two separate baselines act as a target geometry for the SACET wing to achieve. The first baseline features a bell-shaped lift distribution achieved by only changes in local twist, as shown in Figure 8. This model has a fixed symmetrical airfoil with no added camber. The second wing features an identical twist distribution, with a uniform camber along the entire span. The camber is equivalent to a traditional flap deflected at 10 degrees; however, the airfoil is instead a smooth, continuous shape. This cambered Eppler 479 airfoil was traced from a physical profile previously created from the morphing wing by Pabon et al. [10]. These baseline wings were created to compare how the SACET wing imitates varying geometries.



Figure 11. Baseline Rigid Models Without and With Camber

C. Numerical Investigation with FlightStream®

FlightStream® [18] was employed to numerically investigate two wing configurations that were tested in the University of Dayton Low-Speed Wind Tunnel. FlightStream® features two primary potential flow solvers: a pressure-based solver and a vorticity solver. For this study, the pressure-based solver was selected due to its strong agreement with wind tunnel experimental results demonstrated in previous studies [10]. This solver utilizes the pressure fields

computed on the model geometry to determine aerodynamic loads. The solution employs a combination of source and sink panels, along with doublet panels at the trailing edge, to solve the potential flow equations. The boundary layer was assumed to be fully turbulent, with viscous coupling enabled to improve the accuracy of the results. A structured mesh, as well as an illustration of the wings' twist distributions for the rear, is illustrated in Figure 12. The structured mesh applied to the wing platform consists of 160 node points spanwise and 80 node points chordwise on both the upper and lower surfaces. These wings represent the non-twisted morphing wing and the target spanwise twist distribution. Both wings imitate the scale of the physical model with an 8 in chord and 16 in half-span, but in a full-span model with symmetry at the root as annotated.

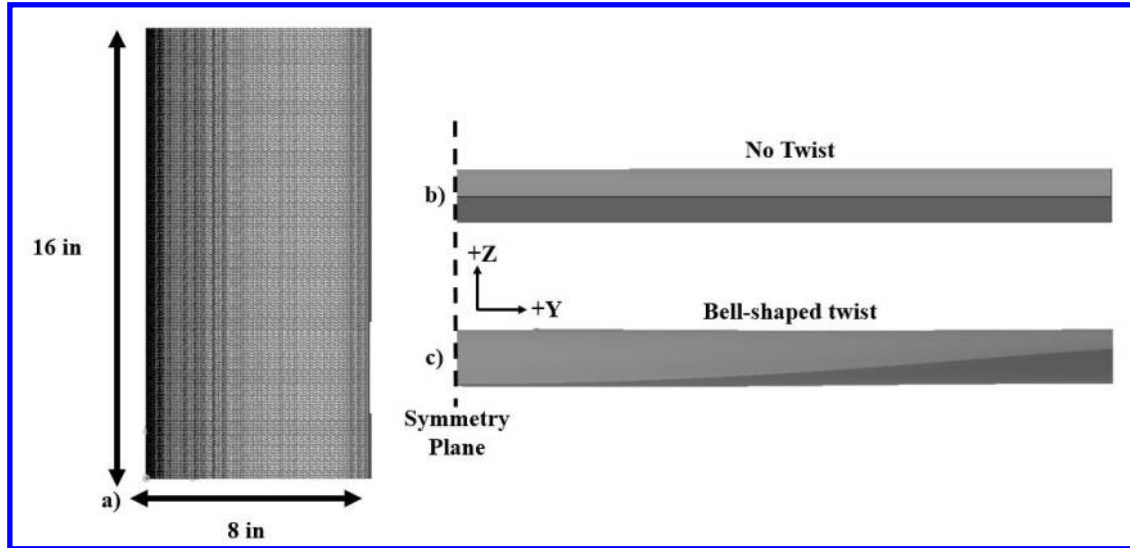


Figure 12 a) Illustration of structured mesh applied to wing platform, (b) & (c) Rear views of twist distribution for FlightStream tested wings.

IV. Results

Experimental testing was conducted to validate the current performance of the morphing wing. The goal was to identify the ideal results for a wing of the same dimensions with a bell-shaped lift distribution. This is produced by the baseline rigid cases. The morphing wing is then tested to compare the results and find if the trends match that of the rigid model. A performance match would then mean the morphing wing is successful in achieving a large change in lift-distribution and vice versa.

A. Experimental Results

Only the SACET wing with rib-embedded servos was tested in the wind tunnel and the resultant coefficients are discussed in this section. Baselines models serve as the ideal performance benchmark for the morphing wing. Figure 13 summarizes the lift behavior of the SACET morphing wing and the corresponding rigid reference wings. Figure 13a compares the morphing wing with no actuation to the same wing with twist actuation only and no camber. The two C_L - α curves lie essentially on top of each other, and any small differences fall within the measurement uncertainty. This indicates that the intended twist distribution was not realized in the experiment. The resistance of the TPU skin to shear and the limited servo torque prevented the ribs from reaching the commanded twist, so the effective spanwise loading remained close to that of the untwisted configuration. This means the current morphing wing design is unable to achieve a substantial performance change with only twist.

Figure 13b illustrates the comparison between the ideal baseline performance and morphing wing only with twist. The twisted rigid wing shows a lower C_L at a given angle of attack than the untwisted case, consistent with the inboard shift in lift expected from the bell-shaped distribution. If the morphing wing had achieved the same twist distribution, its C_L - α curve would coincide with that of the twisted rigid wing. Instead, the morphing wing with twist actuation is

similar to the untwisted wing. The performance not overlapping is consistent with idea that the desired twist distribution was not achieved.

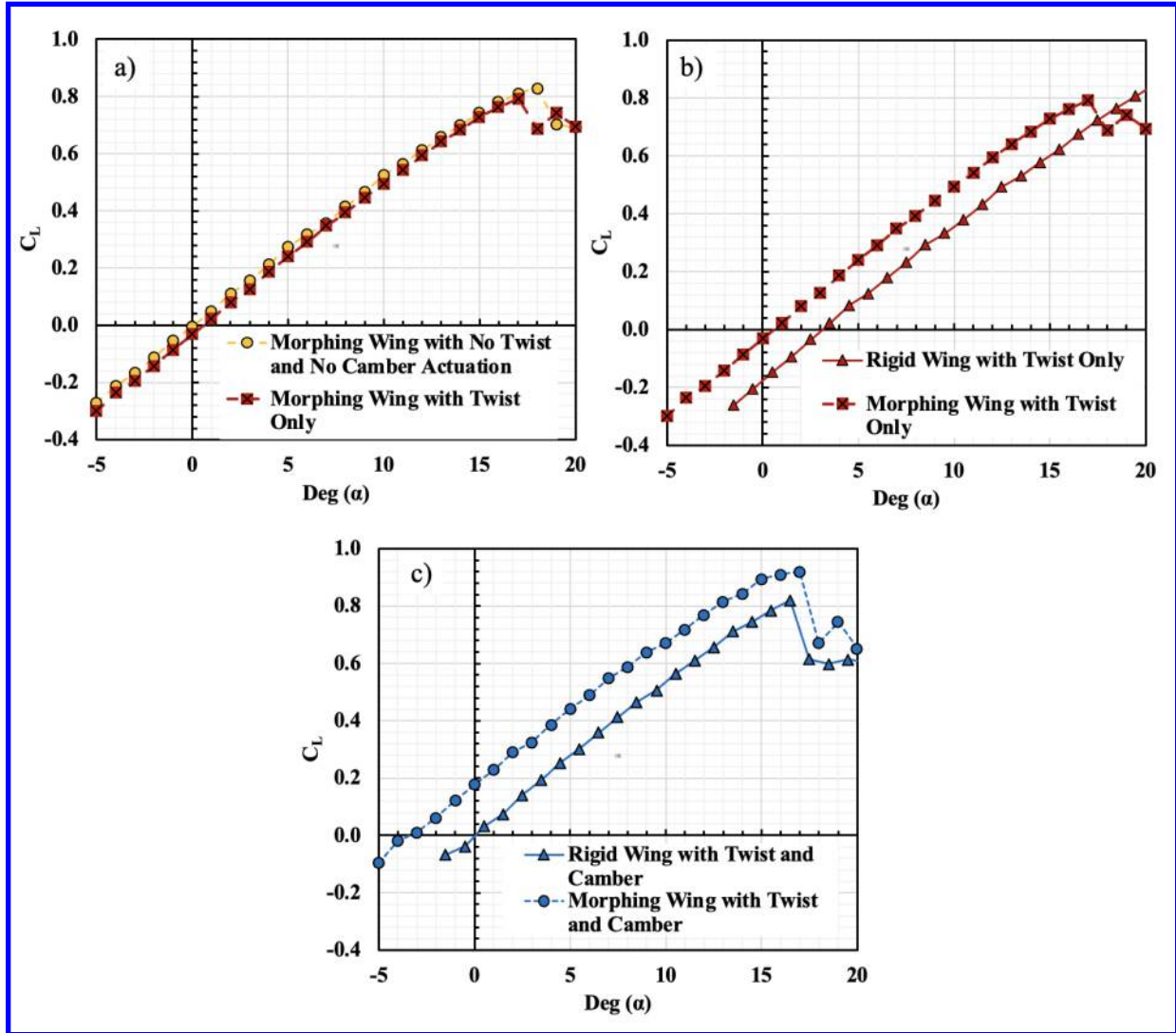


Figure 13 Lift coefficient versus angle of attack for baseline and actuated configurations (a) Morphing wing with no actuation and with twist only (b) Rigid wing with prescribed twist and morphing wing with twist only and (c) Rigid wing with twist and camber and morphing wing with twist and camber.

Figure 13c evaluates the camber mechanism by comparing a baseline rigid wing with both twist and camber to the morphing wing with twist and camber actuated. Adding camber to the twisted rigid wing increases C_L at a given angle of attack by an amount equivalent to a shift of roughly three degrees in α . The morphing wing with camber actuation exhibits a similar three degree offset relative to its own twist only curve, which shows that the camber mechanism is functioning as intended. The remaining discrepancy between the rigid and morphing curves is therefore attributed to the incomplete realization of the bell-shaped twist distribution rather than a deficiency in camber actuation. It is worth noting that applying a non-uniform camber to attempt the bell-shaped lift distribution would face the same limitation from TPU deflection.

After examining the lift behavior, the next step is to assess how the drag coefficient varies with angle of attack for the same configurations. Figure 14 shows the drag performance with the same comparisons as Figure 13. Figure 14a compares the morphing wing with no actuation to the same wing with twist actuation only and no camber. Over most of the angle of attack range, the two C_D versus α curves align closely. Small differences at low and moderate angles

fall within the experimental uncertainty. At the highest angle of attack, the twisted case shows a small drag increase, which is consistent with the slight change in stall behavior seen in the lift data. Overall, the absence of a clear drag change between the configurations again indicates that the intended bell-shaped twist distribution was not achieved by the morphing wing.

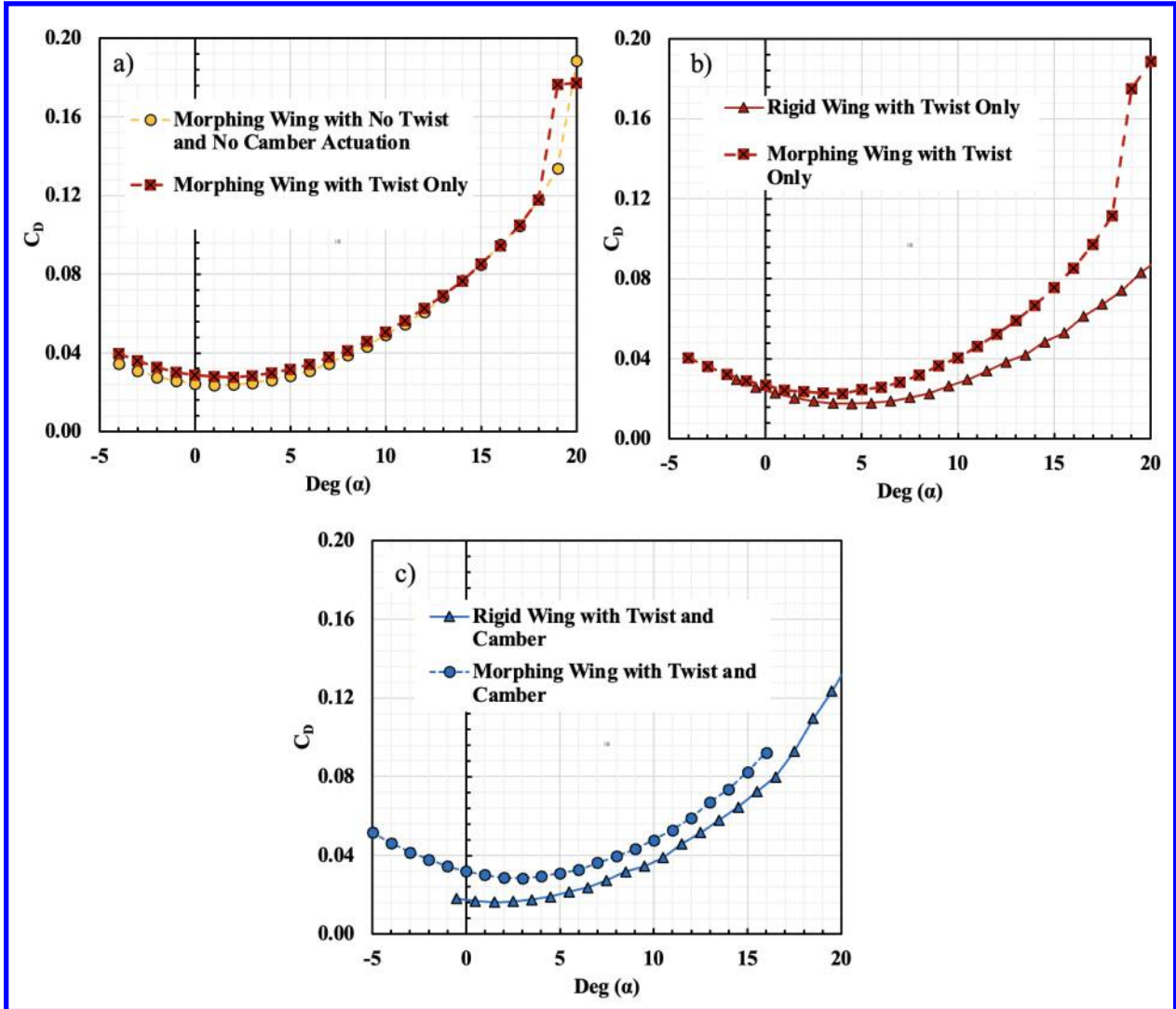


Figure 14 Drag coefficient versus angle of attack for baseline and actuated configurations (a) morphing wing with no actuation and with twist only (b) rigid wing with prescribed twist and morphing wing with twist only and (c) rigid wing with twist and camber and morphing wing with twist and camber.

Figure 14b compares the morphing wing with twist actuation to the rigid wing that has the prescribed bell-shaped twist. The rigid twisted wing exhibits a lower C_D over much of the angle range. This is expected performance associated with the bell-shaped distribution and consistent with the lower C_L seen in Figure 13. Because the rigid wing simply represents the ideal bell-shaped twist distribution, the key point is that the morphing wing should match its C_D versus α curve if the commanded twist were fully realized. The fact that the morphing curve does not coincide with the rigid one confirms that the achieved twist was smaller than the design value. The added surface roughness from the TPU patches could have also contributed to the higher drag seen in Figure 14b.

Figure 14c evaluates the effect of camber by comparing a rigid wing with twist and camber to the morphing wing with both twist and camber actuated. Adding camber increases drag for both configurations, especially at higher angles of attack, which agrees with the higher lift levels seen earlier. Across most of the range, the morphing wing shows a

drag penalty of about 0.01 to 0.02 in C_D relative to the rigid reference. This nearly uniform offset suggests that the camber mechanism adds some extra profile and skin friction drag but otherwise functions as intended. Taken together with the lift results, these trends support the conclusion that the primary performance limitation of the current SACET prototype is the incomplete realization of the twist distribution rather than a lack of camber authority.

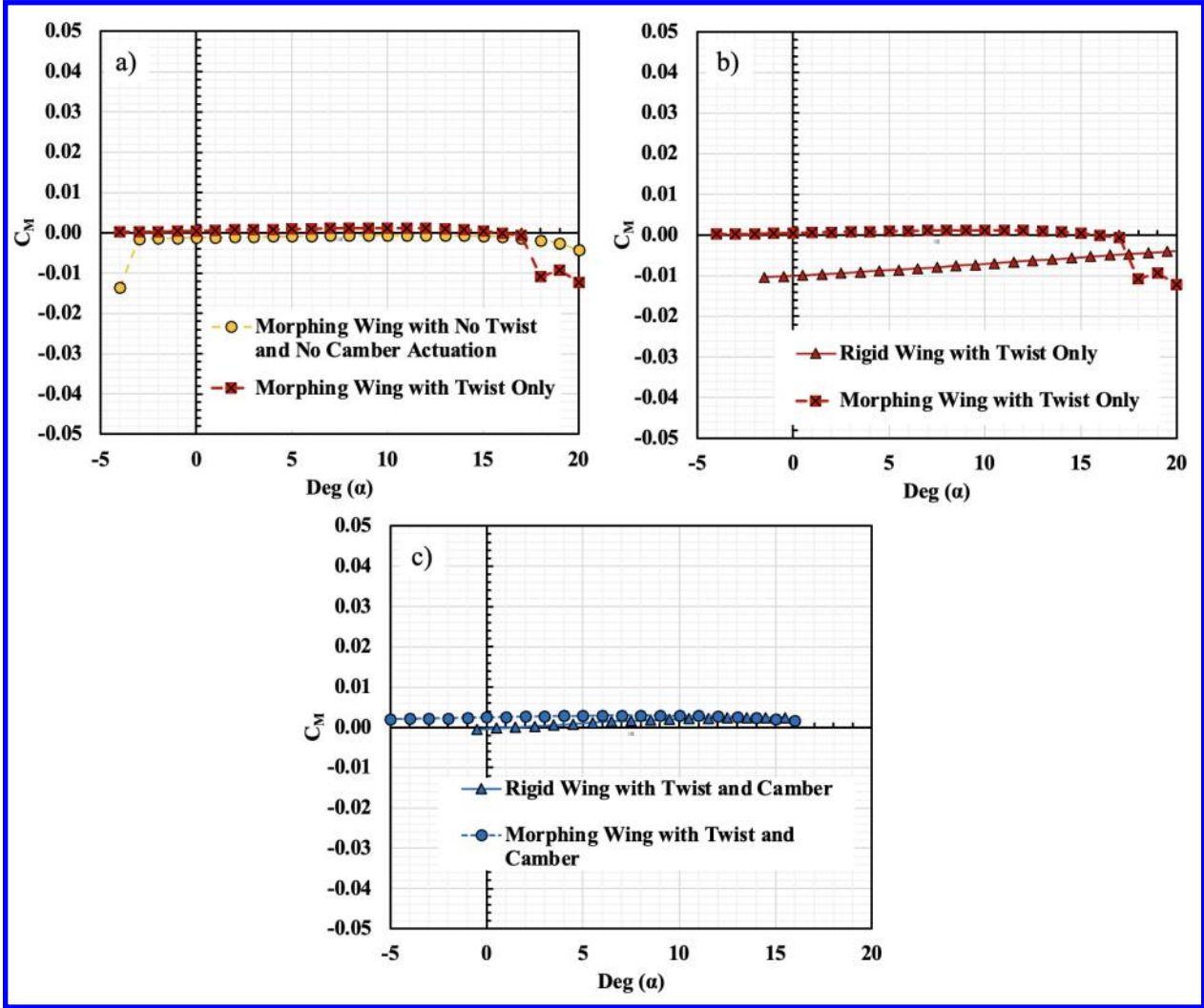


Figure 15 Quarter chord pitching moment coefficient versus angle of attack for baseline and actuated configurations (a) morphing wing with no actuation and with twist only (b) rigid wing with prescribed twist and morphing wing with twist only and (c) rigid wing with twist and camber and morphing wing with twist and camber.

Next the pitching moment coefficient C_M is compared between the same cases as before. The pitching moment behavior provides an additional check on whether the intended twist and camber distributions were achieved. For a symmetric wing referenced about the quarter chord, the moment coefficient should be close to zero across angle of attack. Figure 15a confirms this for the baseline configuration. The morphing wing with no actuation and the morphing wing with twist actuation only both exhibit nearly zero C_M at the quarter chord over the tested range. Because twist should shift the spanwise loading and move the effective center of pressure, the pitching moment for the twisted case is expected to deviate from zero. The fact that the measured C_M remains unchanged therefore indicates that the commanded twist was not realized.

Figure 15b shows the ideal response through the rigid wing with the prescribed bell-shaped twist. In this case C_L is reduced and the center of pressure moves aft, which results in a more negative quarter chord moment that stays roughly

constant with angle of attack. The rigid twisted wing therefore illustrates the expected nose down pitching moment associated with the bell-shaped lift distribution. If the morphing wing had achieved the same twist, its C_M curve would match the rigid one. Instead, the morphing wing with twist actuation remains clustered near zero, which again confirms that the effective twist is much smaller than the design value.

Figure 15c examines the combined effect of twist and camber. For the rigid wing, adding camber to the twisted configuration brings the quarter chord moment back toward zero, consistent with the higher lift and forward shift in center of pressure. The morphing wing with twist and camber actuation shows a similar trend, with a slight increase in C_M relative to its twist only case. This modest change demonstrates that the camber mechanism does influence the pitching moment as expected, while the remaining differences between rigid and morphing curves are mainly due to the incomplete realization of the twist distribution.

B. Numerical FlightStream® Results

As mentioned earlier, the goal for the morphing wing design is to achieve a bell-shaped lift distribution that is known to suppress tip vortex features [15, 16]. A panel method simulation was conducted on the wing with an uniform distribution and a bell-shaped twist distribution to show the differences in the wake structure for the two wings. Three angles of attack were examined at 3° , 8.5° , and 12° . The design angle for the twisted wing is 8.5° at $C_L = 0.3$. The resulting angles were chosen to show the wake behavior before, during, and after the design angle of attack. Shown below in Figure 16 is a rear view of the wake for both wings at all tested angles of attack. A contour is applied to the streamlines representing the vertical velocity component.

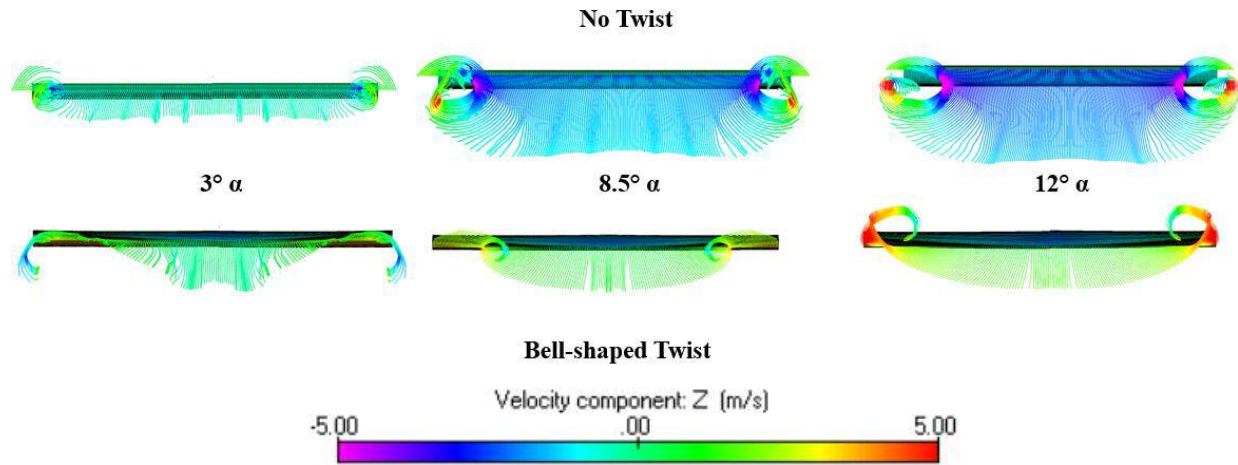


Figure 16. Rear view of the wake at 10 x/c downstream for both tested wings at $\alpha = 3^\circ$, 8.5° , and 12° . The top row presents the results for the untwisted wing configuration, while the bottom row shows the corresponding wake for the wing incorporating a bell-shaped twist distribution.

Figure 16 shows the wake produced by a wing with no twist and another with bell-shaped twist as seen from a view downstream. The no twist model shows the expected strong wing tip vortex at all three angles. Alternatively, the bell-shaped twist model presents a unique wake, particularly at 8.5° . At 3° the bell-shaped twist shows a weak wingtip vortex with that rotates in clockwise direction (opposite to conventional rotation of counter-clockwise). Increasing the angle of attack to the design angle, the bell-shape twist produces an inboard shifted vortex, also known as the wing vortex. This characteristic wing vortex marks the transition between the downwash and upwash going from the root to tip seen by Pabon et. al [14]. More importantly, at this angle of attack, the tip vortex is suppressed. Further increasing the angle of attack past the design condition, the bell-shaped twist now produces a conventional wingtip vortex.

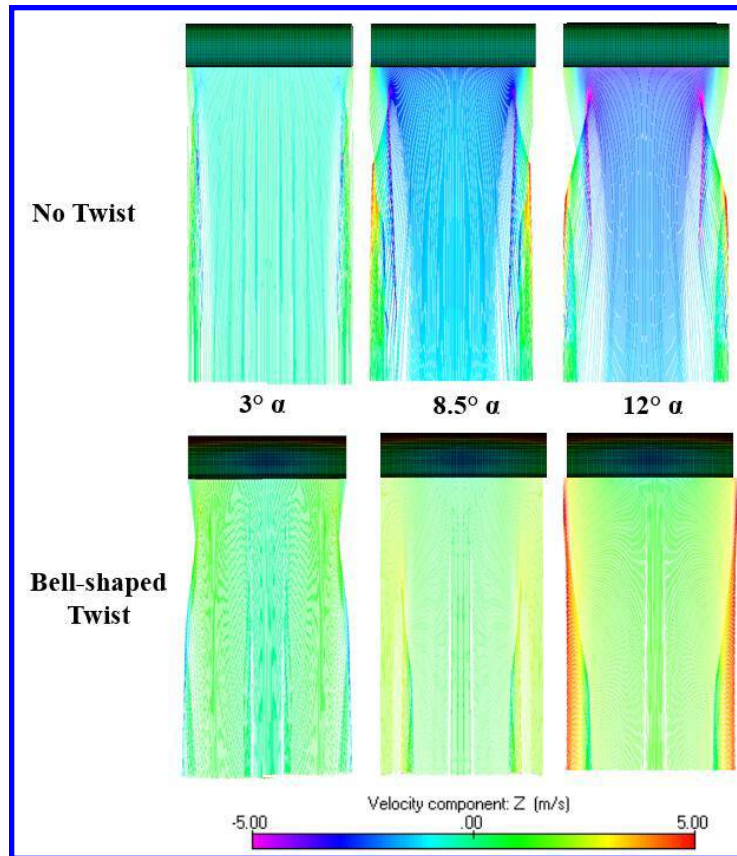


Figure 17 Top view of the wake for both tested wings at 10 x/c downstream for $\alpha = 3^\circ$, 8.5° , and 12° .

Figure 17 shows the same near wake from top. This view allows for the roll up of the wing to more easily be seen. The no twist model produces tighter roll up representing a stronger vortex core compared to the bell-shaped twist at all three angles of attack. At $\alpha = 3^\circ$, a delay in rollup can be seen, with the bell-shaped twist wings presenting the onset of a vortex much later than the non-twisted case. Increasing to $\alpha = 8.5^\circ$, the bell-shaped twisted wing presents a nearly 2-dimensional shape until $\sim 7 x/c$ at its design condition. Overall, the numerical results show that large changes in flow field can be achieved by altering the local lift distribution along the span. Transition between the two would allow for optimization of the wing characteristics depending on flight conditions.

V. Conclusion

This work presented a dual-mode morphing wing capable of camber and twist actuation with the goal of mimicking a bell-shaped lift distribution. Three SACET design concepts were designed, and one was fabricated and tested in the University of Dayton Low Speed Wind Tunnel. Designs included a rib embedded servo design, a gear train driven twist shaft, and a linkage based single rod concept. All three mechanisms were designed using simple geometric models that relate actuator motion to local rib twist and were sized to match the target twist distribution that produces a bell-shaped lift profile at the design condition. Only the rib embedded servo SACET wing was fabricated and tested in the University of Dayton Low Speed Wind Tunnel and compared with rigid wings that embodied the ideal twist and camber distributions.

The experimental results showed that the current prototype did not achieve the commanded twist. The lift, drag, and moment curves for the twist only case were nearly identical to the baseline configuration, and the quarter chord moment remained near zero, which confirms that the spanwise loading did not shift as intended. In contrast, the rigid wing with prescribed twist yielded the expected reduction in lift, drag, and moment. Camber actuation functioned in changing the C_L at all angles. The morphing wing produced an effective three-degree shift in the lift curve that matched the rigid reference, and the moment change showed the same trend. Overall, the SACET concept shows potential for

future bell-shaped lift implementations. The present results indicate that improved twist authority, reduced skin resistance, and refined internal mechanics are needed before the full aerodynamic benefits of combined camber and twist morphing can be realized.

VI. Future Work

Experimental testing showed the current capability of the SACET wing design and its shortcomings. While camber actuation was successful in altering the C_L , the twist actuation was unable to provide a substantial change. Several challenges that hinder the design have been identified that need to be addressed in the next iterations. Most prominent is the resistance to deflection at the trailing edge. This location shows the most resistance because it is furthest from the center of rotation. A location further from the center of rotation must deflect more compared to a closer location. This produces a higher shear load on the material, which the current TPU patch does not allow for. Despite being a flexible material, the print orientation going from root to tip means this shear force goes between layer lines, where elastic properties are at their worst. Another challenge left to address is the reduction in design complexity. Although including additional servos to allow for independent adjustment of each rib allows for more adjustment, it also introduces more complexity and weight. The consideration of a simpler design, such as the discussed SACET with linkages, could be a compromise between functionality and simplicity. In future studies, the focus is on incorporating a skin more conducive to actuation. Current investigations include alternate materials and skin fabrication.

VII. Acknowledgements

The authors would like to acknowledge the following University of Dayton Senior Design Team Students for designing the SACET with Linkage wing shown in this paper: Abigail Boyer, Jacob Harding, Gage Harestad, Samuel Parlett, and Brennan Strickler.

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